

AAMOD BHATT

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SUMMARY

Machine Learning Engineer with an accepted IEEE conference paper on real-time video inpainting, a merged open-source contribution setting a 1.1179 bits-per-byte record on the parameter-golf language-modeling benchmark, and 50+ LoRA/QLoRA fine-tuning experiments. Builds and ships ML systems end to end using PyTorch, HuggingFace Transformers, and multi-GPU cloud infrastructure.

EDUCATION

Visvesvaraya Technological University - Acharya Institute of Technology

B.E. in Artificial Intelligence and Machine Learning | CGPA: 9.28/10

Bangalore, India

Sept 2022 - May 2026

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Artificial Intelligence, Machine Learning, Computer Vision, Database Management Systems

RESEARCH EXPERIENCE

Flow-Guided Real-Time Video Inpainting - IEEE CONECCT 2026 (Accepted)

Bangalore, India

Kavitha Nair R., Aamod Bhatt, et al., Proc. 12th IEEE Intl. Conf. on Electronics, Computing and Communication Technologies

- Co-authored a flow-guided latent diffusion model (FGLDM) for real-time video inpainting with temporally consistent reconstruction across frames.
- Paper accepted for publication at IEEE CONECCT 2026; camera-ready version submitted.

Open-Source Contribution - Parameter Golf Language-Modeling Benchmark

2026

- Set a leaderboard record of 1.1179 bits per byte by applying the Muon optimizer for test-time training; pull request reviewed and merged into the upstream open-source repository.
- Ran distributed training experiments on 8x NVIDIA H100 GPUs using self-funded cloud compute.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript

Machine Learning: PyTorch, HuggingFace Transformers, LoRA/QLoRA (PEFT), Diffusion Models, OpenCV, LangChain, LangGraph

Full Stack: React, Next.js, Node.js, Express, Tailwind CSS, REST APIs, PostgreSQL

Tools: Git, Docker, Google Cloud Platform, AWS EC2, Linux

EXPERIENCE

Vitalcep (D2C Health Supplement Brand)

Dec 2024 - Jan 2026

Growth & Automation Engineer (Part-Time)

- Built an end-to-end Meta Ads automation pipeline (creative, campaign, and optimization stages) using n8n on AWS EC2, reducing campaign setup time by 85%.
- Automated tracking across 40+ ad sets, improving ROAS by 2.3x and reducing daily manual work from 4 hours to 20 minutes.
- Designed backend workflows handling 10,000+ events per day at 99.5% uptime, integrating PostgreSQL, Telegram, and Google Drive.

PROJECTS

CycleFlow - Consistency-Preserving Video Generation

PyTorch, Diffusion Models, RAFT

- Developing a video generation system using cycle-guided flow matching and latent alignment for temporal consistency.
- Combines RAFT optical-flow priors with latent diffusion models to preserve coherent motion across frames.

ChainScope - LLM Reasoning-Chain Diagnostics

Python, Open Source

- Built an open-source diagnostic tool that traces and analyzes large language model reasoning chains to localize failure modes.

FTunePrompt - Fine-Tuning vs. Prompting Framework

PyTorch, PEFT

- Conducted 50+ controlled experiments comparing LoRA fine-tuning against prompting across tasks and data scales; built a CLI tool that recommends the optimal adaptation strategy.

TLDRUN - ML Paper-to-Code System

LangChain, Python

- Converts machine learning papers into runnable starter repositories, generating dataset loaders, configs, and training pipelines.

Recursive Knowledge Engine - Retrieval and Reasoning

Python, RAG

- Built a recursive retrieval pipeline with iterative query refinement for multi-hop question answering.